05/03/2021

Code-A

Aakash

Medical|IIT-JEE|Foundations

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MM : 720

NEET BOOSTER TEST SERIES (NBTS) for NEET-2021

Time : 3.00 Hrs.

Test - I2

Topics Covered: Complete Syllabus of Class-XI & XII (NEET Pattern)

Instructions :

(i) Use blue/black ballpoint pen only to darken the appropriate circle.

(ii) Mark should be dark and should completely fill the circle.

(iii) Dark only one circle for each entry.

(iv) Dark the circle in the space provided only.

(v) Rough work must not be done on the Answer sheet and do not use **white-fluid** or any other **rubbing material** on Answer sheet.

(vi) Each question carries 4 marks. For every wrong response 1 mark shall be deducted from total score.

PHYSICS

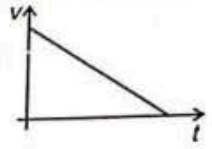
Choose the correct answer :

1. If a physical quantity is given by $x = \frac{a^2 b^3}{c\sqrt{d}}$ and the percentage errors of measurement in a, b, c and d are respectively 2%, 1%, 3% and 4%. Maximum percentage error in the value of x will be

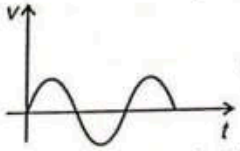
(1) 8% (2) 3%

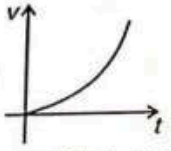
(3) 12% (4) 18%

2. Which of the following curve can't represent motion in one dimension?

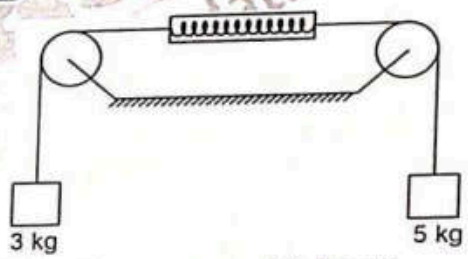
(1) 

(2) 

(3) 

(4) 

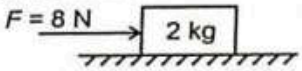
3. If all the pulleys and strings are ideal, then the reading of spring balance in given situation is (g = 10 m/s^2)



(1) 3 kg wt (2) 5 kg wt

(3) 8 kg wt (4) 3.75 kg wt

4. A block of mass 2 kg is placed on the floor. The coefficient of static friction between the mass and floor is 0.6. A force F is acting on it as shown in the figure. The frictional force on the block is



(1) Zero (2) 12 N

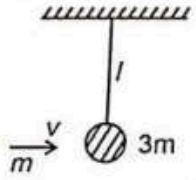
(3) 2 N (4) 8 N

5. A ball is dropped from a height of 100 m. Due to collision with the floor it loses its 20% kinetic energy. After collision ball will rise to a height of

(1) 80 m (2) 20 m

(3) 40 m (4) 50 m

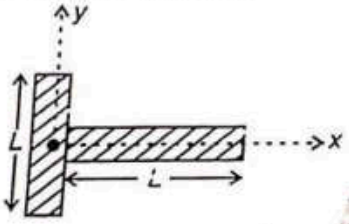
6. A bullet of mass m moving with velocity v horizontally strikes another object of mass $3m$ suspended vertically with a thread as shown. If bullets remain embedded in the object then find the minimum value of v so that composite body can completes its vertical circular path.



(1) $\sqrt{5gl}$ (2) $\sqrt{16gl}$

(3) $\sqrt{20gl}$ (4) $\sqrt{80gl}$

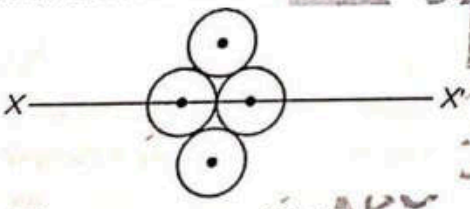
7. A 'T' shaped rod is placed in xy plane as shown in the figure. If the rod is uniform and both parts are of equal length L and mass m then distance of centre of mass of the system from origin will be (Ignore thickness of the rod)



(1) $\frac{L}{2}$ (2) $\frac{L}{4}$

(3) $4\sqrt{2}$ (4) $\frac{3L}{4}$

8. The moment of inertia of a solid sphere about its diameter is I. Four such spheres are arranged as shown in the figure. The moment of inertia of the system about the axis XX' will be



(1) $13I$ (2) $15I$

(3) $17I$ (4) $19I$

9. A satellite of mass $\frac{M}{8}$ is revolving around the planet of mass M in a circular orbit of radius R. Then its angular momentum is

(1) $\sqrt{\frac{GM^3R}{4}}$ (2) $\sqrt{\frac{GM^3R}{64}}$

(3) $\sqrt{\frac{GM^3R}{32}}$ (4) $\sqrt{\frac{GM^3R}{8}}$

10. The motion of a particle is expressed by the equation $a = -bx$, where a is the acceleration and x is the displacement from the equilibrium position and b is constant.

Time period of the particle will be

(1) $2\pi\sqrt{b}$ (2) $2\pi / b$

(3) $2\pi / \sqrt{b}$ (4) $\sqrt{2\pi / b}$

11. If two waves represented by $y_1 = 4\sin(\omega t + kx)$ and $y_2 = 3\sin(\omega t + kx + \pi/3)$ interfere at a point, then amplitude of the resulting wave will be closest to

(1) 7 (2) 5

(3) 6 (4) 11

12. An earthquake generates both transverse (S) and longitudinal (P) sound waves in the earth. The speed of the S waves is about 4.5 km/s and that of P waves is about 8.0 km/s. A Seismograph records P and S waves from an earthquake. The first wave P arrives 4 minutes before the first S wave. The epicentre of the earthquake is located at a distance about

(1) 25 km (2) 1200 km

(3) 800 km (4) 2500 km

13. A cable that can support a maximum load of 240 kg is cut into two equal parts. The maximum load that can be supported by either part is

(1) 60 kg (2) 120 kg

(3) 240 kg (4) 480 kg

14. A liquid A of relative density 3.36 is filled in a U-tube up to a length of 10 cm height, which contains mercury. Another liquid B is filled in other arm of U-tube to a height of 8 cm. If upper levels of A and B are same then density of liquid B is ($\rho_{Hg} = 13.6$ g/cm^3)

(1) 1.6 g/cm^3 (2) 0.6 g/cm^3

(3) 1.36 g/cm^3 (4) 0.8 g/cm^3

15. Two Carnot engines X and Y are operated in succession. The first engine X receives heat from a source at $T_1 = 800$ K and rejects to sink at T_2 K. The second engine Y receives heat rejected by the first engine and rejects to another sink at $T_3 = 300$ K. If the work output of two engines are equal, then the value of T_2 is

(1) 550 K (2) 460 K

(3) 1100 K (4) 1600 K

16. A uniform electric field of 40 N/C exists along the x-axis in a space. The potential difference $(V_B - V_A)$ for the points A(4 m, 3 m) and B(8 m, 6 m) will be

(1) 120 V (2) 160 V

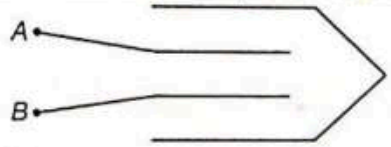
(3) −160 V (4) −120 V

17. A charge Q is uniformly distributed over a large square plate of copper. The electric field at a point very close to the centre of the plate is 40 V/m. If the copper plate is replaced by a paper plate of identical dimensions and carrying the same charge Q uniformly distributed, then the electric field at the point P will be

(1) 80 V/m (2) 40 V/m

(3) 20 V/m (4) 10 V/m

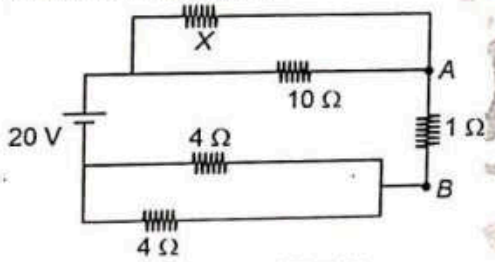
18. If capacitance between two successive plates in given arrangement is C, then equivalent capacitance between the points A and B, will be



(1) $\frac{3C}{2}$ (2) $\frac{2C}{3}$

(3) $2C$ (4) $\frac{C}{2}$

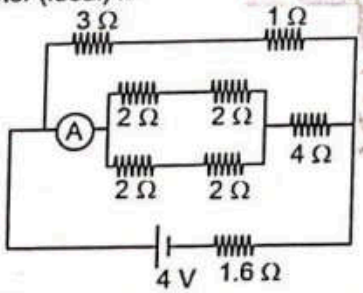
19. An electrical circuit is given in the figure. Power developed in the 1 ohm resistance will be maximum if value of X is



(1) 10 Ω (2) 4 Ω

(3) 1 Ω (4) Zero

20. In the given circuit, the value of reading of ammeter (ideal) is



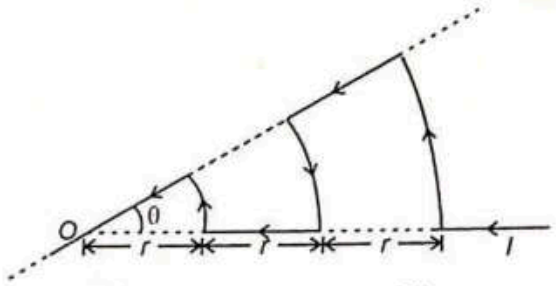
(1) 0.6 A

(2) 1 A

(3) 0.4 A

(4) 1.4 A

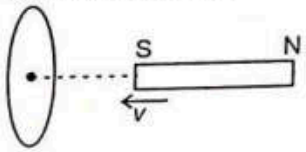
21. A conductor carrying current I is bent in the shape as shown in the figure. Common centre of all the three arcs is O. Magnetic field intensity at the point O is



(1) $\frac{\mu_0 I\theta}{12\pi r}$ (2) $\frac{\mu_0 2I\theta}{4\pi r}$

(3) $\frac{\mu_0 I\theta}{24\pi r}$ (4) $\frac{5\mu_0 I\theta}{24\pi r}$

22. In the figure given below the magnet is moved towards the stationary coil with speed v and induced emf in the coil is ε. Now, if magnet and coil both recede away from one another, each moving with speed $2v$ (Same position), then induced emf in the coil will be



(1) $\frac{\varepsilon}{4}$

(2) $\frac{\varepsilon}{2}$

(3) 4ε

(4) ε

23. In an AC circuit, current is given by the function $I = 50 \sin (100\pi t)$. The time required for the current to achieve its peak value from $t = 0$, will be

(1) $\frac{1}{400}$s

(2) $\frac{1}{200}$s

(3) $\frac{1}{100}$s

(4) $\frac{1}{50}$s

24. An electromagnetic pulse having intensity of 1200 W/m^2 is incident on a metal surface. If pulse after normal incidence over the plate reflects back completely, then force applied by pulse on the metal surface will be (Given surface area of metal is 2 m^2)

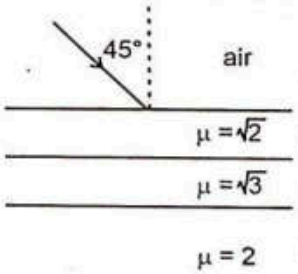
(1) 0.8×10^{-5} N

(2) 0.8×10^{5} N

(3) 1.6×10^{5} N

(4) 1.6×10^{-5} N

25. A light ray is incident on the composite slab as shown in the figure. Angle made by the light ray with the normal in medium 3 will be



(1) $\sin^{-1}\left(\frac{1}{\sqrt{3}}\right)$ (2) $\sin^{-1}\left(\frac{1}{\sqrt{6}}\right)$

(3) $\sin^{-1}\left(\frac{2}{\sqrt{3}}\right)$ (4) $\sin^{-1}\left(\frac{\sqrt{3}}{2}\right)$

26. If the de Broglie wavelength of an electron in the first Bohr's orbit be λ then the minimum radial distance between the electrons in the first and second Bohr's orbit is

(1) $\frac{\lambda}{2}$ (2) $\frac{\lambda}{\pi}$

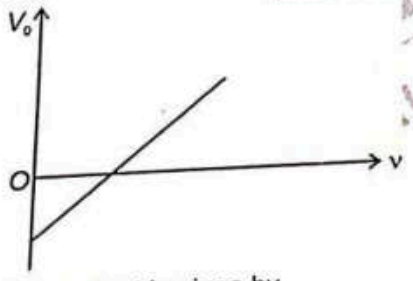
(3) $2\pi\lambda$ (4) $\frac{\lambda}{2\pi}$

27. The half life of ^{60}Co is approximately 5.25 years. In a sample containing 1 g of freshly prepared ^{60}Co, how much of the isotope will be left after 21 years?

(1) 16 mg (2) 31.25 mg

(3) 62.5 mg (4) 6.25 mg

28. The stopping potential V_0, for an experiment of photoelectric effect from a metal surface, is plotted with frequency ν of incident light as shown in the figure.



Planck's constant is given by

(1) Slope of the line divided by the charge on an electron

(2) Slope of the line multiplied by the charge on an electron

(3) Intercept cut by the line on *y*-axis

(4) Product of intercept on *y*-axis and charge of the electron

29. A particle in a certain conservative force field has a potential energy given by $U = \frac{10xz}{y}$. The force exerted on it is

(1) $-\frac{10z}{y}\hat{i} - \frac{10xz}{y^2}\hat{j} - \frac{10x}{y}\hat{k}$

(2) $\frac{10x}{y}\hat{i} - \frac{10z}{y^2}\hat{j} + \frac{10yx}{y^2}\hat{k}$

(3) $-\frac{10z}{y}\hat{i} + \frac{10xz}{y^2}\hat{j} - \frac{10x}{y}\hat{k}$

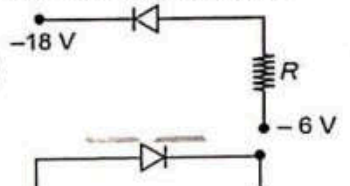
(4) $\frac{10z}{y}\hat{i} + \frac{10xz}{y^2}\hat{j} + \frac{10x}{y}\hat{k}$

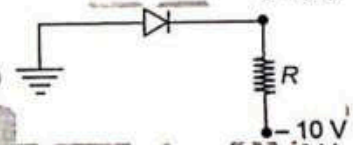
30. In an NPN transistor the collector current is 24 mA. If 80% of electrons reach collector, its emitter current in mA is

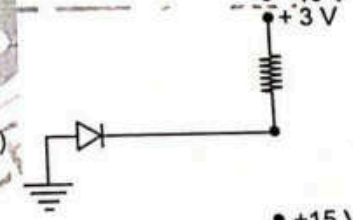
(1) 36 (2) 30

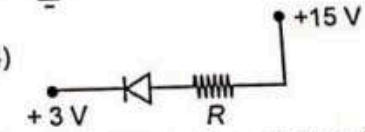
(3) 24 (4) 64

31. Of the diodes shown in the following diagrams, which one is reversed biased?

(1)


(2)


(3)


(4)


32. The total energy of an electron in its nth excited state of hydrogen atom is *K*. Its kinetic energy in this state is

(1) $-K$ (2) $K/2$

(3) $2K$ (4) $\frac{2K}{3}$

33. A man weighing 60 kg stands on a weighing scale in a lift which is moving upwards with a uniform acceleration of 6 m/s². Reading on the weighing scale will be (g = 10 m/s²)

(1) 24 kg (2) 60 kg

(3) 96 kg (4) 70 kg

34. A circular coil, carrying a certain current produces a magnetic field B_0 at its centre. The coil is now rewound so as to have 3 turns and the same current is passed through it. The new magnetic field at the centre is

(1) $B_0/9$ (2) $9B_0$

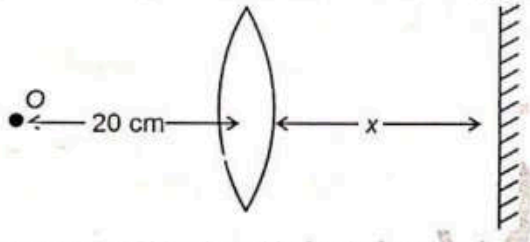
(3) $B_0/3$ (4) $3B_0$

35. A coil in the shape of an equilateral triangle of side 'L' is suspended between the pole pieces of a permanent magnet such that magnetic field $\vec{B}$ is in the plane of the coil. If due to a current I in the triangle a torque τ acts on it, the side L of the triangle is

(1) $\frac{4\tau}{\sqrt{3}IB}$ (2) $\frac{2\tau}{\sqrt{3}IB}$

(3) $\sqrt{\frac{2\tau}{\sqrt{3}IB}}$ (4) $2\sqrt{\frac{\tau}{\sqrt{3}IB}}$

36. A convex lens of focal length $f = 20$ cm is placed in front of a plane mirror as shown in the figure.

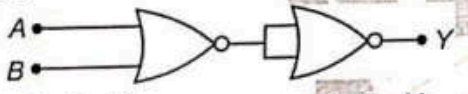


An object 'O' is placed in front of convex lens. For what value of x final image will coincides with the object?

(1) 20 cm (2) 40 cm

(3) 30 cm (4) All of the above

37. In the following circuit, the output Y for all possible inputs A and B is expressed by the truth table



(1)

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

(2)

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

(3)

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

(4)

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

38. Two coherent monochromatic light beams of intensities I and $4I$ are superimposed. The maximum and minimum possible intensities in the resulting beam are

(1) $5I$ and I (2) $3I$ and I

(3) $9I$ and I (4) $9I$ and $3I$

39. The sum of magnitudes of two forces acting at a point is 16 N and their resultant is $8\sqrt{3}$ N at 90° with the force of smaller magnitude. The two forces (in N) are

(1) 8 N and 6 N (2) 4 N and 10 N

(3) 3 N and 11 N (4) 2 N and 14 N

40. Initial temperature and volume of a gas are T and V respectively. If gas expands to 27 V following the equation VP^3 = constant, then final temperature of the gas will become

(1) $T/27$ (2) $9T$

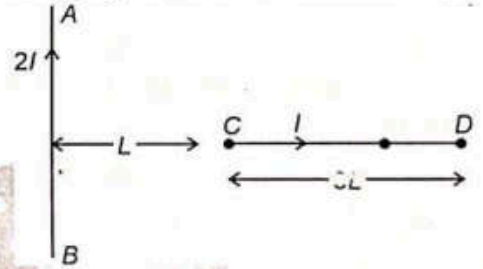
(3) $27T$ (4) $\frac{T}{9}$

41. For a certain organ pipe, three successive resonance frequencies are observed at 425, 595 and 765 Hz respectively. Taking the speed of sound in air to be 340 ms^{-1}, the length of the organ pipe is

(1) 3 m (2) 2 m

(3) 4 m (4) 1 m

42. A conductor of length $3L$ carrying a current I is placed perpendicular to a long straight conductor AB carrying current $2I$ as shown in the figure.



The magnitude of force acting on AB is

(1) $\frac{\mu_0}{4\pi} I^2 \ln 4$

(2) $\frac{\mu_0}{8\pi} 4I^2 \ln 2$

(3) $\frac{\mu_0}{2\pi} I^2 \ln 2$

(4) $\frac{2\mu_0}{\pi} I^2 \ln 2$

43. Electrons in a certain energy level $n = n_1$, can emit 3 spectral lines. When they are in another energy level $n = n_2$ they can emit 6 spectral lines. The ratio of orbital speeds of electrons in the two orbits are in the ratio

(1) $\frac{3}{4}$ (2) $\frac{4}{3}$

(3) $\frac{16}{9}$ (4) $\frac{9}{16}$

44. A charge Q is distributed over two concentric hollow spheres of radii r and R ($> r$) such that the surface densities are equal. The potential at the common centre is

(1) $\frac{Q}{4\pi(r+R)}$ (2) $\frac{Q(R+r)}{4\pi\varepsilon_0(R^2+r^2)}$

(3) $\frac{Q(R^2+r^2)}{4\pi\varepsilon_0(R+r)^2}$ (4) $\frac{Q}{4\pi\varepsilon_0 R}$

45. A concave lens of focal length 10 cm is placed in a liquid such that refractive index of lens and liquid is same. The new focal length of the lens will be

(1) 0

(2) ∞

(3) f

(4) $f/2$

CHEMISTRY

46. Number of atoms in 0.2 g molecule of O_3 is

(1) N_A (2) 0.2 N_A

(3) 0.6 N_A (4) 1.2 N_A

47. Orbital in which electron density is not along the axis, is

(1) $2p_x$ (2) $3p_y$

(3) $4p_z$ (4) $3d_{xy}$

48. Element of highest electronegativity among the following is

(1) O (2) S

(3) Se (4) Te

49. Which molecule does not have tetrahedral geometry?

(1) CH_4 (2) NH_4^+

(3) PCl_5 (4) CCl_4

50. Paramagnetic species among the following is

(1) H_2 (2) N_2

(3) F_2 (4) O_2^+

51. Density (in g/L) of Ozone at 1 atm and 207°C is

(1) 12.2 (2) 1.22

(3) 10.2 (4) 22.2

52. 1 mol of an ideal gas reversibly compressed from 5 L to 0.5 L at 127°C temperature. Work done involves in this process is

(1) 921.2 R (2) –921.2 R

(3) – 460.6 R (4) 460.6 R

53. EAN of metal ion in complex ion $\left[Co(CO)_6\right]^+$ is

(1) 27 (2) 38

(3) 36 (4) 37

54. Most acidic compound among the following is

(1) $HCCl_2COOH$ (2) FCH_2COOH

(3) $ClCH_2COOH$ (4) CH_3COOH

55. Standard enthalpy of formation is zero for

(1) White phosphorous (2) Diamond

(3) Hydrogen liquid (4) Iodine liquid

56. Correct order of solubility of salts Ag_2CO_3, Ag_2CrO_4 and Ag_2SO_4 is

(Given : K_{sp} (Ag_2SO_4) > K_{sp} (Ag_2CO_3) > K_{sp} (Ag_2CrO_4))

(1) $Ag_2SO_4 > Ag_2CrO_4 > Ag_2CO_3$

(2) $Ag_2SO_4 > Ag_2CO_3 > Ag_2CrO_4$

(3) $Ag_2CO_3 > Ag_2CrO_4 > Ag_2SO_4$

(4) $Ag_2CO_3 > Ag_2SO_4 > Ag_2CrO_4$

57. Salt involving anionic hydrolysis, is

(1) CH_3COONa (2) NH_4Cl

(3) $(NH_4)_2SO_4$ (4) $C_6H_5\overset{+}{N}H_3Cl^-$

58. Equimolar aqueous solution of which of the following species has highest hydrogen ion concentration at 298 K? [Given : K_a (HNO_2) > K_a (CH_3COOH) > K_a (HCN) > K_a (C_6H_5OH)]

(1) HNO_2 (2) CH_3COOH

(3) C_6H_5OH (4) HCN

59. Select the reaction which is not redox among the following.

(1) $TiCl_4(l) + 2Mg(s) \longrightarrow Ti(s) + 2MgCl_2(s)$

(2) $Cr_2O_3(s) + 2Al(s) \longrightarrow Al_2O_3(s) + 2Cr(s)$

(3) $2H_2O(l) + F_2(g) \longrightarrow 4HF(aq) + O_2(g)$

(4) $AgNO_3(aq) + KI(aq) \longrightarrow AgI(s) + KNO_3(aq)$

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60. E° of cell reaction,

$Cu(s) + I_2(s) \longrightarrow Cu^{2+}(aq) + 2I^-(aq)$, is

$\left(E^\circ_{Cu^{2+}/Cu} = 0.34\ V \text{ and } E^\circ_{I_2/I^-} = 0.54\ V\right)$

(1) –0.2 V (2) 0.88 V

(3) 0.2 V (4) 0.14 V

61. Strength of 2.8 volume solution of hydrogen peroxide is

(1) 2% (2) 3%

(3) 0.85% (4) 1.7%

62. Alkali metal of highest density among the following is

(1) Li (2) K

(3) Rb (4) Cs

63. Select the correct statement among the following.

(1) LiCl is deliquescent and crystallises as a hydrate of $LiCl \cdot 2H_2O$

(2) Lithium forms ethynide on reaction with ethyne

(3) LiF is much more soluble in water than LiBr

(4) Lithium is softest among alkali metals

64. Incorrect match of name of compounds is

(1) Plaster of paris : $CaSO_4 \cdot \frac{1}{2} H_2O$

(2) Gypsum : $CaSO_4 \cdot H_2O$

(3) Bleaching powder : $CaOCl_2$

(4) Tricalcium silicate : Ca_3SiO_5

65. Diborane on hydrolysis by water gives

(1) Boric oxide + Oxygen gas

(2) Borazine + Oxygen gas

(3) Boric acid + Hydrogen gas

(4) Boric oxide + Hydrogen gas

66. Which species is least likely to exist?

(1) SiF_6^{2-} (2) $GeCl_6^{2-}$

(3) $[Sn(OH)_6]^{2-}$ (4) $SiBr_6^{2-}$

67. Oxide, which is distinctly acidic among the following is

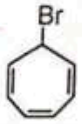
(1) CO (2) GeO

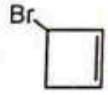
(3) SnO (4) PbO_2

68. Select the molecule which will give fastest precipitation reaction with $AgNO_3$ solution?

(1) 

(2) 

(3) 

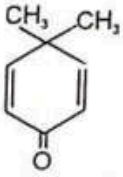
(4) 

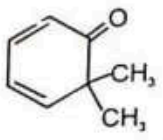
69. In the Kjeldahl's method for estimation of nitrogen present in a soil sample, ammonia evolved from 0.5 g of sample neutralised 20 ml of 1 M HCl. The percentage of nitrogen in the soil is

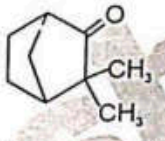
(1) 14% (2) 28%

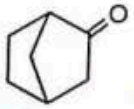
(3) 56% (4) 50%

70. Which among the following can exhibit tautomerism?

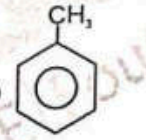
(1) 

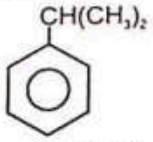
(2) 

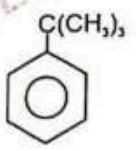
(3) 

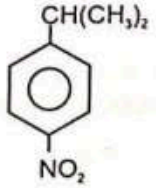
(4) 

71. Which one of the following is most reactive towards electrophilic aromatic substitution?

(1) 

(2) 

(3) 

(4) 

72. Excessive concentration of nitrate in drinking water mainly causes

(1) Kidney damage

(2) Liver damage

(3) Methemoglobinemia

(4) Laxative effect

73. Ferrimagnetic substance among the following is

(1) MnO (2) Fe_3O_4

(3) C_6H_6 (4) CrO_2

74. The vapour pressure of pure liquids A and B respectively are 900 and 800 mmHg. Partial vapour pressure of B in the liquid mixture in which A and B are mixed in the ratio of 2 : 3 is
(1) 300 mmHg
(2) 480 mmHg
(3) 840 mmHg
(4) 360 mmHg

75. Reduction potential of hydrogen electrode which is placed in a solution of pH = 4, is
(1) 0.1 V (2) –0.24 V
(3) –0.56 V (4) –0.48 V

76. Select the ion having maximum limiting molar conductivity (in water) at 298 K among the following.
(1) Na^+ (2) K^+
(3) OH^- (4) Br^-

77. Rate constant (in mol L^{-1} s^{-1}) of a zero Order reaction in which reactant concentration changes from 0.2 M to 0.05 M in 20 sec is
(1) 7.5 (2) 0.75
(3) 0.0075 (4) 0.075

78. Maximum coagulating power for coagulation of congo red sol is of
(1) Al^{3+} (2) PO_4^{3-}
(3) Ba^{2+} (4) SO_4^{2-}

79. Vapour phase refining is used for refining of
(1) Zr (2) Fe
(3) Zn (4) Cu

80. In acid solution on disproportionation of HNO_2, oxide of nitrogen formed is
(1) NO_2
(2) N_2O_3
(3) N_2O
(4) NO

81. Reaction of aq. $ZnSO_4$ with aq. NH_4OH gives
(1) White precipitate
(2) Brown precipitate
(3) Deep Blue solution
(4) Black precipitate

82. Number of P=O bond(s) present in cyclotrimetaphosphoric acid is
(1) Zero (2) 1
(3) 3 (4) 6

83. The element of highest melting point in second transition series is
(1) Zr (2) Nb
(3) Mo (4) Rh

84. Which among the following has highest number of unpaired electron(s)?
(Atomic number : Pr = 59, Gd = 64, Er = 68 and Yb – 70)
(1) Pr (2) Gd
(3) Er (4) Yb

85. Optically active molecule among the following is
(1) $[Cr(H_2O)_4Cl_2]Cl$
(2) fac - $[Co(H_2O)_3Cl_3]$
(3) trans - $[Cr(en)_2Cl_2]Cl$
(4) cis - $[Co(en)_2Cl_2]Cl$

86. Which one is most reactive towards S_N1 reaction?
(1) CH_3CH_2Br (2) $(CH_3)_2CHBr$
(3) $(C_6H_5)_3CBr$ (4) $(C_6H_5)_3CCH_2Br$

87. Alcohol which gives immediate turbidity with Lucas reagent, is
(1) CH_3OH (2) CH_3CH_2OH
(3) $(CH_3)_2CHOH$ (4) $(CH_3)_3COH$

88. Tollen's test is not given by
(1) HCOOH (2) CH_3CH_2CHO
(3) CH_3COCH_3 (4) $PhCH_2CHO$

89. Which amine does not react with Hinsberg's reagent?
(1) CH_3NH_2 (2) $PhCH_2NH_2$
(3) $(CH_3)_3N$ (4) $(Ph)_2CHNH_2$

90. Select the correct statement among the following.
(1) Nylon 6, 6 is an addition polymer
(2) Polyvinyl is a thermosetting polymer
(3) Dilute aqueous solution of boric acid is used as a weak antiseptic for eyes
(4) Sodium lauryl sulphate is non-ionic detergent

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BOTANY

91. Mango and potato belong to same
(1) Family (2) Class
(3) Genera (4) Order

92. The taxonomic aid 'Key' is based on the set of contrasting characters. Each character in this set is called
(1) Lead (2) Couplet
(3) Element (4) Fact

93. In five kingdom classification, *Chlamydomonas* is placed in kingdom
(1) Monera (2) Plantae
(3) Animalia (4) Protista

94. The following features are related to a fungus :
- Karyogamy is delayed after plasmogamy.
- Meiospores are formed inside the ascus.

One of the fungi that exhibits the above given features is
(1) *Penicillium* (2) *Albugo*
(3) *Agaricus* (4) *Puccinia*

95. Match the following columns and select the **correct** option.

	Column-I		Column-II
a.	Pulvinus	(i)	Green coloured leaf-like petiole
b.	Lamina	(ii)	Expanded part of leaf
c.	Stipule	(iii)	Swollen leaf base
d.	Phyllode	(iv)	Leaf like structure at the leaf base

(1) a(iv), b(ii), c(i), d(iii) (2) a(iii), b(ii), c(iv), d(i)
(3) a(iv), b(i), c(iii), d(ii) (4) a(iii), b(ii), c(i), d(iv)

96. Floral formula of a plant is given below :

$\oplus \sigma\!\!\varphi\ \overline{P_{(3+3)}\ A}_{3+3}\ \underline{G}_{(3)}$

The floral formula is **not** indicating that
(1) Androecium is epiphyllous
(2) Gynoecium is syncarpous
(3) Flower is bisexual
(4) Placentation is axile

97. The band like structures in the endodermis of roots are impermeable to water due to the presence of
(1) Lignin (2) Pectin
(3) Suberin (4) Chitin

98. Select the **incorrect** statement from the following.
(1) In spring season, cambium produces vessels that have narrow lumens
(2) In winter, fewer xylary elements are formed by cambium
(3) Early wood is formed during spring season
(4) Autumn wood is darker in colour and has a higher density

99. Consider the following chart :

Classes of algae	Major pigments	Stored food
Chlorophyceae	A	–
Phaeophyceae	B	C
Rhodophyceae	D	–

Select the option which is **true** for A, B, C or D.
(1) A - Chlorophyll *a*, chlorophyll *d* and phycocyanin
(2) B - Chlorophyll *a*, chlorophyll *c* and fucoxanthin
(3) C - Floridean starch
(4) D - Chlorophyll *a*, Chlorophyll *b* and xanthophyll

100. In which of the following aspects, *Cycas* is different from *Pinus*?
(1) Dependency of female gametophyte on parent sporophyte
(2) Possessing tap root system
(3) Bearing seeds which are not enclosed within fruit wall
(4) Forming symbiotic association with nitrogen fixing cyanobacteria in coralloid roots

101. The envelope in bacteria that allows them to hide from host's immune system is made up of
(1) Peptidoglycans (2) Lipoproteins
(3) Fatty acids (4) Polysaccharides

102. The chromosomes that have satellite
(1) Lack primary constriction
(2) Have secondary constriction
(3) Do not undergo replication
(4) Are always telocentric

103. "The cell does **not** replicate DNA but remains metabolically active". This statement is true for which of the following stages?

(a) G_2 phase (b) S phase

(c) G_0 phase (d) G_1 phase

The **correct** one(s) is/are

(1) All except (b) (2) Only (d)

(3) Only (a) and (d) (4) Only (c) and (d)

104. Match the following columns and select the **correct** option.

	Column-I		Column-II
a.	Metaphase I	(i)	Centrioles replicate
b.	Prophase II	(ii)	Formation of double metaphasic plate
c.	Interkinesis	(iii)	Centromere splits
d.	Anaphase II	(iv)	Nucleolus disintegrates

(1) a(ii), b(i), c(iv), d(iii)

(2) a(iii), b(i), c(iv), d(ii)

(3) a(ii), b(iv), c(iii), d(i)

(4) a(ii), b(iv), c(i), d(iii)

105. Which one of the following is **not** true regarding the events responsible for the opening of stomata?

(1) Dissociation of malic acid into malate and H^+ ions

(2) Transport of K^+ ions into the guard cells

(3) Increase of CO_2 concentration in substomatal cavity

(4) Formation of potassium malate in the guard cells

106. Mineral elements are translocated through xylem in

(1) Both inorganic and organic forms

(2) Only in inorganic forms

(3) Only in organic forms

(4) Only in ionic forms

107. Which of the following is **not** considered as essential mineral nutrient for plant?

(1) Nitrogen (2) Phosphorous

(3) Potassium (4) Carbon

108. Deficiency symptoms of Mg, Fe and Ca in plants are also seen due to toxicity of

(1) Zn (2) Mn

(3) Mo (4) Cu

109. According to chemiosmotic hypothesis, the ultimate event that provide energy for the synthesis of ATP in chloroplast is

(1) Breakdown of proton gradient across the thylakoid membrane

(2) Transfer of protons from stroma to the lumen of thylakoid

(3) Uphill transport of electrons to the photosystem

(4) Removal of protons from stroma by reduction of $NADP^+$ to $NADPH + H^+$

110. Carbon dioxide in the atmosphere is considered as a major limiting factor influencing the rate of photosynthesis because

(1) Carbon dioxide is the main source of carbon for plants

(2) The level of carbon dioxide in the atmosphere is far below for optimum photosynthesis

(3) Carbon dioxide is not easily dissolved in the water

(4) Carbon dioxide is absorbed by the plants only through stomata

111. In respiration, which of the following substrates on its complete oxidation will result maximum net gain of ATP molecules?

(1) One molecule of glucose

(2) One molecule of fructose-6-phosphate

(3) Two molecules of glyceraldehyde-3-phosphate

(4) Two molecules of pyruvic acid

112. The complex in ETS of mitochondria which has two copper centres is

(1) Complex II (2) Complex IV

(3) Complex I (4) Complex III

113. Two leaves (A and B) of a plant have sizes 10 cm^2 and 20 cm^2 respectively. After one day their sizes become 20 cm^2 and 30 cm^2 respectively.

On the basis of given information select the **correct** statement.

(1) Both the leaves show same relative growth rate

(2) Relative growth rate of leaf A is more than that of leaf B

(3) Relative growth rate of leaf B is more than that of leaf A

(4) Absolute growth rate of leaf A is more than that of leaf B

114. Match the following columns and select the **correct** option.

	Column-I		Column-II
a.	Gibberellin	(i)	Counteracts apical dominance
b.	Ethylene	(ii)	Precursor is acetyl-CoA
c.	Abscisic acid	(iii)	Inhibits seed germination
d.	Cytokinin	(iv)	Initiates flowering in mango tree

(1) a(iv), b(ii), c(i), d(iii)
(2) a(ii), b(iii), c(i), d(iv)
(3) a(iii), b(iv), c(ii), d(i)
(4) a(ii), b(iv), c(iii), d(i)

115. Which of the following statements is **not** true w.r.t. asexual reproduction?
(1) It is a quick method of reproduction
(2) It can occur through unspecialised parts of parents
(3) It occurs only in lower plants and animals
(4) It can occur with or without gamete formation

116. All the following are related with *Spirogyra* **except** one. Identify that one.
(1) Formation of motile male gamete
(2) Internal fertilisation
(3) Presence of chlorophyll *a* and *b*
(4) Vegetative reproduction by fragmentation

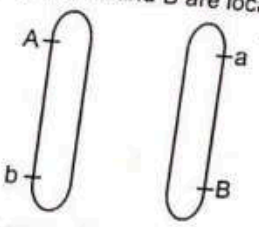
117. The ovary has generally single ovule in
(1) Wheat (2) Papaya
(3) Orchid (4) Water melon

118. Select the **incorrect** match from the following.

(1)	Funicle	–	Stalk of ovule
(2)	Chalaza	–	Basal part of ovule
(3)	Synergids	–	Have filiform apparatus
(4)	Megaspore mother cell	–	Haploid

119. A woman with blood group AB and carrier of colour blindness is married with a normal man whose blood group is O. What would be probability of their B blood group children to be colourblind?
(1) 6.25% (2) 50%
(3) 12.5% (4) 25%

120. In the diploid cells of a flowering plant, alleles of two linked genes A and B are located as follows.



Out of 800 generative cells of the pollen grains formed by this plant, 50 has both dominant alleles A and B on the same chromosome as shown below.



What should be the distance between these two genes in this plant?
(1) 50 cM
(2) 25 cM
(3) 12.5 cM
(4) 6.25 cM

121. Read the following statements and choose the option which is true for them?
Statement-1 : The individual inflicted with Turner's syndrome is a sterile male.
Statement-2 : In human when a normal egg fuses with an abnormal sperm that has both the sex chromosomes, it results in the disorder called Klinefelter's syndrome.
(1) Both the statements are correct
(2) Both the statements are incorrect
(3) Only statement-1 is correct
(4) Only statement-2 is correct

122. In eukaryotes, transcription of rRNAs is catalysed by
(1) Only RNA polymerase I
(2) Both RNA polymerase I and II
(3) Both RNA polymerase II and III
(4) Both RNA polymerase I and III

123. In which of the following organisms, there is single origin of replication?
(1) *Vicia faba*
(2) *Saccharomyces cerevisiae*
(3) *Escherichia coli*
(4) *Aspergillus niger*

124. tRNA with which of the following anticodon is **not** found?

(1) 

(2) 

(3) 

(4) 

125. In *lac* operon, the genes *lac a* and *i*-gene respectively code for
(1) β-galactosidase and permease
(2) Transacetylase and β-galactosidase
(3) Transacetylase and repressor protein
(4) Permease and repressor protein

126. The crucial step for the success of breeding experiment is
(1) Selection and testing of superior recombinants
(2) Release and commercialisation of new cultivars
(3) Evaluation and selection of parents
(4) Cross hybridisation among the selected parents

127. For which work, Ernest chain and Howard Florey were awarded Nobel prize?
(1) Establishing full potential of penicillin as an effective antibiotic
(2) Extracting the antibiotic streptomycin from a species of *Streptomyces*
(3) Exploring the various benefits of *Saccharomyces*
(4) Finding the use of various microorganisms for human welfare.

128. A bioactive molecule used as an immunosuppressive agent in organ-transplant patient is produced by
(1) *Monascus purpureus*
(2) *Trichoderma polysporum*
(3) *Aspergillus niger*
(4) *Geotrichum candidum*

129. Select the **incorrect** statement from the following
(1) A stage of suspended development found in many zooplanktons is called diapause
(2) Most of the lower vertebrates are able to maintain homeostasis by physiological means
(3) Nearly all plants cannot maintain a constant internal environment
(4) Snail and fishes escape in time by the mechanism called aestivation

130. *Paramecium caudatum* and *paramecium aurelia* in same culture will show
(1) Competitive co-existence
(2) Parasitism
(3) Commensalism
(4) Competitive exclusion

131. A secondary consumer in an ecosystem
(1) Is a herbivorous animal
(2) Occupies the second trophic level
(3) Is said to be a primary carnivore
(4) Can occupy the first trophic level

132. A transitional community in xerarch succession does **not** include
(1) Mosses
(2) Annual grasses
(3) Lichens
(4) Shrubs

133. Which of the following is **not** among the Evil Quartet that cause biodiversity loss?
(1) Over exploitation
(2) Biomagnification
(3) Habitat loss and fragmentation
(4) Alien species invasions

134. How many of the following are *ex-situ* conservation strategies?

Botanical garden, National park, Sacred groves, Seed bank, Aquaria, Wildlife safari park, Wildlife sanctuary

(1) Five (2) Three
(3) Four (4) Two

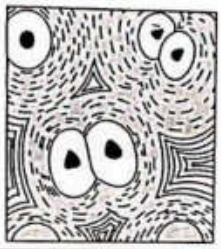
135. The correct option regarding the contribution of various greenhouse gases to global warming in increasing order is
(1) $CFCs < NO_2 < CO_2 < CH_4$
(2) $NO_2 < CFCs < CH_4 < CO_2$
(3) $NO_2 < CH_4 < CO_2 < CFCs$
(4) $CFCs < N_2O < CH_4 < CO_2$

Medicaliit - Foundations

ZOOLOGY

136. Ciliated epithelium is present in
(1) PCT of cortical nephrons
(2) Intestinal mucosa
(3) Inner lining of bronchi
(4) Inner lining of buccal cavity

137. Identify the tissue shown in the figure given below and choose the option which is **correctly** matched with its characteristics.



(1)	Blood	–	It is main circulatory fluid in the body for transportation of various substances
(2)	Bone	–	Having hard and non-pliable matrix due to presence of calcium
(3)	Cartilage	–	Material in between cells is solid and pliable and resists compression
(4)	Dense connective tissue	–	Collagen fibres are present in rows between many parallel bundles of fibres

138. Match column-I with column-II and choose the option containing only **correct** match.

	Column-I		Column-II
A.	Tight junctions	p	Perform cementing to keep neighbouring cells together
B.	Adhering junctions	q	Helps to stop substances from leaking across a tissue
C.	Gap junctions	r	Facilitate the cells to communicate with each other by connecting the cytoplasm of adjoining cells

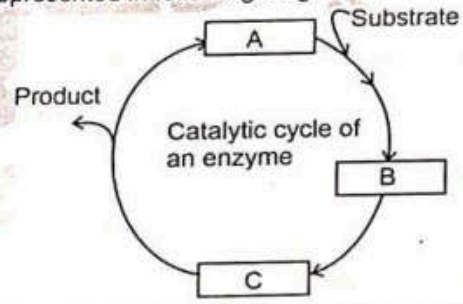
(1) A(p), B(q), C(r)
(2) A(q), B(r), C(p)
(3) A(r), B(p), C(q)
(4) A(q), B(p), C(r)

139. During chemical analysis of living tissue lipids are obtained in acid insoluble fraction known as
(1) Filtrate
(2) Retentate
(3) Substrate
(4) Precipitate

140. Choose the **incorrect** match.

(1)	Heparin	–	Protein prevents coagulation of blood
(2)	Antibody	–	Fights infectious agents
(3)	GLUT-4	–	Enables glucose transport into cells
(4)	Insulin	–	Heteropolymer of amino acids and is secreted from pancreas

141. The catalytic cycle of an enzyme action can be represented in following diagram.



Choose the option which **correctly** represents A, B and C.

	A	B	C
(1)	Inhibitor	Enzyme-product complex	Enzyme-substrate complex
(2)	Enzyme	Enzyme-substrate complex	Enzyme-product complex
(3)	Transition state	Enzyme-substrate complex	Enzyme-product complex
(4)	Cofactor	Transition state	Enzyme-substrate complex

142. In chordates, notochord is mesodermally derived rod-like structure formed on the dorsal side during embryonic development which is ventral to
(1) Nerve cord (2) Liver
(3) Heart (4) Intestine

143. Select the phylum where animals exhibit retrograde metamorphosis in terms of symmetry?
(1) Annelida (2) Coelenterata
(3) Echinodermata (4) Ctenophora

144. Complete the analogy :
Locusta : Malpighian tubules : : *Pinctada* : ____
(1) Mantle (2) Foot
(3) Radula (4) Feather like gills

145. Match column-I and column-II, w.r.t. cockroach. Choose the option with all **correct** match.

	Column-I		Column-II
a.	Mandible	(i)	Pseudopenis and titillator
b.	Gizzard	(ii)	Grinding region and incising region
c.	Genital pouch	(iii)	Thick circular muscle and chitinous plates
d.	Left phallomere	(iv)	Genital chamber and vestibulum

(1) a(ii), b(iii), c(iv), d(i)
(2) a(i), b(ii), c(iii), d(iv)
(3) a(iv), b(iii), c(ii), d(i)
(4) a(iii), b(ii), c(iv), d(i)

146. All of the following are present in blood vascular system of *Periplaneta americana* **except**
(1) Blood (2) Muscular heart
(3) Blood capillaries (4) Blood vessels

147. Choose the **correct** statement w.r.t. digestive tract in human.
(1) Alimentary canal starts with oesophagus and opens to exterior through anus
(2) Muscularis layer consists of outer longitudinal and innermost circular muscle layer in all parts of gut
(3) Mucosa forms rugae in stomach and villi in duodenum
(4) Caecum opens into colon guarded by ileo-caecal valve

148. Sphincter of Oddi is present at the opening of ____ in duodenum. Choose the option which fills the blank **correctly**.
(1) Common bile duct
(2) Common hepatic duct
(3) Pancreatic duct
(4) Hepato-pancreatic duct

149. During swallowing, glottis gets covered by a thin flap, formed of
(1) Dense connective tissue
(2) Specialized connective tissue
(3) Areolar connective tissue
(4) Spongy connective tissue

150. Select the **incorrect** match w.r.t. animals, their respiratory organ and phylum in which they are placed

	Animal	Respiratory organ	Phylum
(1)	*Pheretima*	Moist cuticle	Annelida
(2)	*Torpedo*	Gills	Pisces
(3)	*Psittacula*	Lungs	Chordata
(4)	*Laccifer*	Tracheal tube	Arthropoda

151. Total lung capacity of a normal human is represented by equation TLC = X + RV. X can be any of the lung volumes/capacities given below **except**
(1) TV + FRC (2) TV + IRV + ERV
(3) IC + ERV (4) VC

152. Maximum percentage of plasma is formed by
(1) Minerals (2) Proteins
(3) Water (4) Glucose

153. Choose the **incorrect** match.

(1)	Hypertension	–	May affect cerebrum and other vital organs
(2)	Angina pectoris	–	Acute chest pain due to low O_2 reaching to heart muscle
(3)	Congestive heart failure	–	Heart stops beating
(4)	Heart attack	–	Sudden damage of heart muscle by an inadequate blood supply

154. Read the following statements w.r.t. function of the kidney and choose the option which contains only **incorrect** statements.

(a) PCT helps to maintain the pH and ionic balance of the body fluids by absorption of hydrogen and bicarbonate ions.

(b) Conditional reabsorption of Na^+ and water occurs in PCT.

(c) The ascending limb of loop of Henle is impermeable to water but allows only active transport electrolytes.

(d) Collecting duct allows passage of small amounts of urea into the medullary interstitium.

(1) (a) and (b) only (2) (b) and (c) only
(3) (a), (b) and (c) (4) (d) only

155. Which of the following hormone, keeps a check on the renin- angiotensin mechanism?

(1) Renin (2) Aldosterone
(3) ADH (4) ANF

156. Read the following statements A and B and choose the **correct** answer.

A : In skeletal muscle fibres, light bands contain only actin and dark bands contain only myosin.

B : Length of myosin filaments overlapped by actin filaments, represent H-zone.

(1) Both statements A and B are correct
(2) Both statements A and B are incorrect
(3) Only statement A is correct
(4) Only statement B is correct

157. Select the **mismatch**.

(1)	Intervertebral joint	–	Cartilaginous joint
(2)	Suture	–	Fibrous joint
(3)	Joint between atlas and axis	–	Saddle joint
(4)	Knee joint	–	Hinge joint

158. Which of the following is **not** a component of knee-jerk reflex?

(1) Afferent neuron
(2) Part of CNS
(3) Efferent neuron
(4) Interneuron

159. Read the following statements w.r.t. human eye

(a) Photoreceptors in human eye gets depolarised during darkness and hyperpolarised in response to light stimulus.

(b) Rods are very sensitive to light and contribute to daylight vision.

(c) Cornea consists of dense matrix of collagen and is the non vascular portion of the eye.

(d) The photopigments are embedded in the membrane discs of outer segment of rods and cones.

How many of the above statements are **correct**?

(1) 2 (2) 1
(3) 4 (4) 3

160. Match the following hormones (Column-I) with respective diseases caused due to their imbalance in the body (Column-II)

	Column-I		Column-II
a.	Vasopressin	(i)	Gigantism
b.	Thyroxine	(ii)	Cretinism
c.	Growth hormone	(iii)	Diabetes insipidus
d.	Corticoids	(iv)	Addison's disease

(1) a(iv), b(iii), c(ii), d(i)
(2) a(iii), b(ii), c(i), d(iv)
(3) a(i), b(ii), c(iii), d(iv)
(4) a(ii), b(iv), c(i), d(iii)

161. Choose the **incorrect** statement w.r.t. gluco-corticoids.

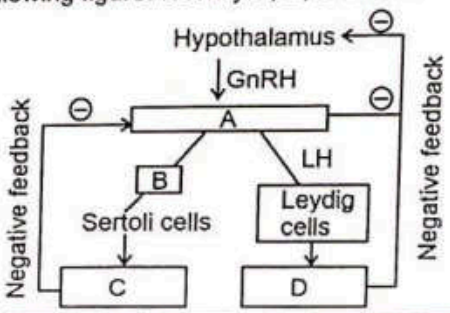
(1) Suppress T cell and B cell response
(2) Stimulate glycogenolysis leading to hyperglycemia
(3) Inhibit cellular uptake and utilization of amino acids
(4) Stimulate the RBC production

162. The females exhibit cyclical changes in the activities of ovary and accessory ducts as well as hormones during the reproductive phase. These cyclical changes occur in

(1) All mammals
(2) Females of placental mammals
(3) Only in mammalian females other than primates
(4) Only in primates

163. Which of the following is a pre-fertilization event?
(1) Gametogenesis (2) Syngamy
(3) Cleavage (4) Embryogenesis

164. Regulation of gonadal function during reproductive phase of a human is given in following figure. Identify A, B, C and D.



	A	B	C	D
(1)	Adenohypophysis	LH	Spermatogenesis	Testosterone
(2)	Anterior pituitary	FSH	Spermatogenesis	Androgens
(3)	Pars-distalis	FSH	Inhibin	Testosterone
(4)	Testis	FSH	Inhibin	Androgens

165. Choose the **incorrect** match.

(1)	Gametogenesis	–	Formation of gametes
(2)	Insemination	–	Transfer of sperms in female genital tract
(3)	Implantation	–	Formation and development of blastocyst
(4)	Gestation	–	Embryonic development

166. Which of the following structures is **not** present in interstitium of testis?
(1) Leydig cells
(2) Sertoli cells
(3) Immunologically competent cells
(4) Blood vessels

167. Read the following statements A and B and choose the **correct** answer.
A : Fertilization in humans is practically feasible only if the ovum and sperms are transported simultaneously to ampulla region of the fallopian tube.
B : Ectopic pregnancies are referred to as pregnancies terminated due to hormonal imbalance.
(1) Both statements A and B are correct
(2) Both statements A and B are incorrect
(3) Only statements B is incorrect
(4) Only statements A is incorrect

168. Which of the following is a hormone releasing IUD?
(1) Lippes loop
(2) Multiload-375
(3) LNG-20
(4) Cu-7

169. Choose the STI which can be cured completely if detected earlier and treated properly?
(1) Hepatitis-B
(2) Genital warts
(3) Genital herpes
(4) HIV

170. Select the **incorrect** match.

(1)	S.L. Miller	–	Created electric discharge in a closed flask
(2)	Alfred Wallace	–	Malay Archipelago
(3)	K.E. von Baer	–	Embryological support for evolution
(4)	Charles Darwin	–	H.M.S. Beagle

171. Lemur and spotted cuscus represent
(1) Convergent evolution
(2) Divergent evolution
(3) Adaptive radiation
(4) Homology

172. All of the following were evolved from thecodonts through geological period **except**
(1) Mammals
(2) Crocodiles
(3) Dinosaurs
(4) Birds

173. Choose the option which fills the blank **correctly** to complete the analogy.

Haemophilus influenzae : Pneumonia :: ____ Malaria

(1) Female *Anopheles*
(2) Female *Culex*
(3) *Plasmodium vivax*
(4) *Wuchereria bancrofti*

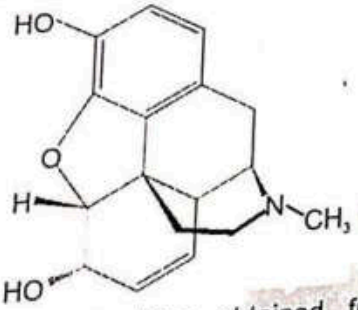
174. B-lymphocytes produce an army of proteins in response to pathogens in our blood to fight against them. This army of proteins is called
(1) Antigens (2) Antibodies
(3) Interferons (4) Enzymes

175. Match column-A with column-B and choose the option with all **correct** match.

	Column-A		Column-B
a.	Asthma	(i)	Infectious disease
b.	Rheumatoid arthritis	(ii)	Infestational disease
c.	AIDS	(iii)	Auto-immune disease
d.	Elephantiasis	(iv)	Allergy

(1) a(i), b(ii), c(iii), d(iv)
(2) a(iii), b(ii), c(i), d(iv)
(3) a(iv), b(iii), c(ii), d(i)
(4) a(iv), b(iii), c(i), d(ii)

176. Select the **correct** statement about the drug given in the following figure



(1) Drug is morphine obtained from *Atropa belladonna*
(2) It binds to cannabinoid receptors present in brain
(3) Its receptors are present in our gastro-intestinal tract
(4) It is a stimulant and activates body functions

177. Which of the following animal breeding is helpful to overcome inbreeding depression?
(1) Inbreeding
(2) Out-crossing
(3) Cross-breeding
(4) Interspecific hybridisation

178. Which of the following options represents the **correct** order of steps in PCR?
(1) Annealing, denaturation, extension
(2) Extension denaturation, annealing
(3) Denaturation, annealing, extension
(4) Denaturation, extension, annealing

179. Read the following statements carefully.
(i) A gene whose expression helps to identify transformed cell is known as selectable marker.
(ii) Separated DNA fragments can be visualized after staining with ethidium bromide only on exposure to UV light
(iii) DNA fragments can be joined to form a recombinant DNA by using restriction enzymes.
(iv) Plasmids are double stranded, transferable, circular DNA having independent replication.

How many of the upper statements is/are **false**?
(1) 2
(2) 3
(3) 4
(4) 1

180. Which of the following proteins is used for treatment of emphysema?
(1) α-lactalbumin
(2) α-1-antitrypsin
(3) Tissue-plasminogen activator
(4) Interferon

□ □ □